

SOFTWARE ENGINEERING LAB 4

Vibe Coding – Chat History

Name	Amar
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Project Name	Real-Time Flappy Bird Clone – Pygame
Lab	Software Engineering Lab – Lab 4
Repository	SETAPESU26/40_flappy-bird

This document contains the questions/prompts used during the Vibe Coding process for the four assigned tasks, along with the corresponding implementation direction and testing outcome.

TASK 1 – REFINE COLLISION DETECTION

USER – QUESTION / PROMPT

Fix the Flappy Bird collision detection. The bird sometimes clips the edge of a pipe without dying, especially at higher speed. Enhance the collision accuracy. Use the bird's full bounding area instead of checking only the center point. Do not change the other game features.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

The collision logic was changed from center-point collision detection to full rectangle collision detection using the bird's bounding rectangle and the pipe rectangles. The implementation used Pygame's **colliderect()** method for both the top and bottom pipes.

USER – TESTING / RESULT

The game was tested by making the bird collide with the edge of a pipe. The collision was detected correctly and the game ended. Task 1 was confirmed working.

TASK 2 – IMPLEMENT GAME OVER CONDITION

USER – QUESTION / PROMPT

Implement a proper Game Over condition. When the bird hits the ground, ceiling, or pipe, stop the gameplay and display a Game Over screen with the final score. Do not add replay, difficulty selection, or sound yet.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

A Game Over screen was added to the game engine. When the bird hits a pipe, the ground, or the ceiling, the game enters the Game Over state. The screen displays **GAME OVER**, the final score, and an exit instruction. Gameplay and normal input are stopped while the Game Over screen is displayed.

USER – TESTING / RESULT

The game was run and a collision produced a visible Game Over screen showing the final score. Task 2 was confirmed working before moving to Task 3.

TASK 3 – ADD REPLAY OPTION AND DIFFICULTY

USER – QUESTION / PROMPT

Add a replay option after Game Over. Allow the player to choose Easy, Medium, or Hard pipe speed/gap, or exit. Use keyboard options: 1 for Easy, 2 for Medium, 3 for Hard, and 4 for Exit. Preserve the existing collision detection and Game Over behavior.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

Difficulty settings were added to the game engine. The selected difficulty controls pipe speed and gap size. The Game Over screen now provides four options: **1 - Easy**, **2 - Medium**, **3 - Hard**, and **4 - Exit**. Selecting 1, 2, or 3 resets the game using the selected settings.

USER – TESTING / RESULT

The replay menu was tested successfully. Easy, Medium, and Hard restart the game with their respective settings, while Exit closes the game. Task 3 was confirmed working.

TASK 4 – ADD SOUND FEEDBACK

USER – QUESTION / PROMPT

Add basic sound feedback to the Flappy Bird game. I need a sound when the bird flaps, a sound when the player passes a pipe and scores, and a sound when the bird dies. Keep all the previous features working and do not break the replay and difficulty functionality.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

Three sound effects were added: **flap.wav**, **score.wav**, and **death.wav**. The GameEngine loads the sounds from the sounds folder. The flap sound plays on Space or mouse click, the score sound plays when the bird successfully passes a pipe, and the death sound plays when the bird hits a pipe, ground, or ceiling. A flag was also used so the death sound plays only once per game.

USER – TESTING / RESULT

The sounds were tested successfully. Flapping, scoring, and death all produced audio feedback. Replay after Game Over was also tested to ensure the sound state resets correctly. Task 4 was confirmed working.

FINAL PROJECT STATUS

Task	Feature	Status
Task 1	Refine Collision Detection	Completed
Task 2	Game Over Condition + Final Score	Completed
Task 3	Replay + Easy/Medium/Hard Difficulty	Completed
Task 4	Flap / Score / Death Sound Feedback	Completed

Final deliverables completed: before-change video, after-change video, updated source code pushed to the personal GitHub repository, and this Vibe Coding chat history document.